

CogniPDF AI

Enterprise Multi-Document Research & Neural Voice Workspace

System Documentation & Technical Guide | Version: 2.4.0 | Stack: Flask / Groq / ChromaDB

1. Overview & What This Application Does

CogniPDF AI is an intelligent document assistant that lets you upload one or multiple PDF documents (such as research papers, user manuals, financial reports, legal contracts, or scanned paper forms) and ask questions about them in natural human language.

Instead of reading through hundreds of pages manually, CogniPDF instantly searches across all your files, pulls out the exact relevant passages, and uses state-of-the-art Artificial Intelligence to write clear, direct answers with page citations. You can type your questions or use your voice, and you can listen to the answers read aloud like an audiobook.

■ **Key Benefit:** Built on a modern **RAG (Retrieval-Augmented Generation)** architecture, the AI never makes up facts or hallucinates. It answers questions strictly grounded in the content of your uploaded PDFs and provides exact clickable page references.

2. How the System Works (End-to-End Flow)

The application processes your documents in six simple, automated stages:

- **Step 1: Document Ingestion (The Eyes):** When you upload a PDF, the app reads the text in sub-milliseconds using **PyMuPDF**. If the document is a scanned image or photo with no digital text, the built-in **RapidOCR** engine scans the image like human eyes and extracts the printed words.
- **Step 2: Slicing into Paragraphs (The Slicer):** An AI cannot read a 500-page book all at once. The app cuts the document into bite-sized chunks (~1,000 characters each, with a 200-character overlap so sentences aren't cut mid-thought). Each piece is labeled with its exact **File Name** and **Page Number**.
- **Step 3: Converting Words to Meaning (The Translator):** The app uses a dense embedding model (**all-MiniLM-L6-v2**) running locally on your computer. It converts each text chunk into a list of mathematical numbers (vectors) that represent the *concept and meaning* of the words.
- **Step 4: Storing in Memory (The Memory Vault):** These numerical representations are stored inside **ChromaDB** (an in-memory vector database). This allows the system to search by topic and meaning (for example, finding 'revenue' when you ask about 'income or profits').
- **Step 5: Reasoning & Synthesizing the Answer (The Brain):** When you ask a question (via text or microphone), the app pulls the **top 5 to 8 most relevant paragraphs** from ChromaDB and passes them to **llama-3.3-70b-versatile on Groq LPUs**. The AI reads those pieces and writes a clear answer with exact citations and 3 smart follow-up suggestions in less than one second.
- **Step 6: Voice Playback (The Voice):** When you click 'Read Aloud', **Microsoft Neural Speech (Edge-TTS)** or **ElevenLabs** turns the written answer into human-like audio and plays it instantly.

3. AI Models & Tools Reference

Here is a simple summary of every model and tool used in the application and its exact purpose:

Role / Model	Category	Provider	Plain English Purpose
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llama-3.3-70b-versatile The Main Brain	Reasoning LLM	Meta / Groq	The primary intelligence engine. Reads retrieved passages, writes accurate answers, generates exact citations, and suggests follow-up questions.
groq/compound-mini Fast Summarizer	Ultra-Fast LLM	Groq	Ultra-low-latency model for instant executive summaries and quick prompt chips.
openai/gpt-oss-120b Flagship Model	Deep Reasoning LLM	Groq	High-capacity model for complex cross-document comparisons and deep technical analysis.
qwen/qwen3.6-27b Math & Multilingual	Specialized LLM	Qwen / Groq	Specialized for mathematical calculations, tables, formulas, and non-English documents.
all-MiniLM-L6-v2 The Meaning Model	Vector Embeddings	Local ONNX	Converts text into 384-dimensional math vectors for semantic search. Runs 100% locally and free with zero API charges.
RapidOCR (PP-OCRv4) Image Reader	Local OCR Engine	ONNX Runtime	Reads printed text from scanned PDFs and photos directly without third-party cloud fees.
en-US-AriaNeural The Voice	Neural TTS	Microsoft	Default natural, expressive voice for reading out AI answers clearly.
ElevenLabs Multilingual Studio Narrator	Voice Cloning	ElevenLabs	Optional hyper-realistic voice personalities (Rachel, Adam, Antoni, Bella) for audiobook-style playback.

4. Technology Stack Breakdown

- **Backend Server:** Python 3.10+, Flask 3.1.3 (RESTful API architecture)
- **Production WSGI Server:** Gunicorn 26.2.0 (configured with --workers 1 --threads 4 --timeout 120 for thread-safe vector state)
- **Concurrency & State Protection:** Python threading.Lock() mutex ensuring reliable uploads and deletions
- **RAG & AI Framework:** LangChain Core, LangChain Groq, and ChromaDB 1.5+ (In-Memory Vector Store)
- **Document Parsing:** PyMuPDF 1.28+ (C-accelerated fast text), pdfplumber 0.11+ (tables), pypdf 6.16+
- **Local OCR:** RapidOCR 1.4+ running on ONNX Runtime (zero cloud fees)
- **Speech Technologies:** edge-tts 7.2+, ElevenLabs API, and browser Web Speech API (Voice Dictation)
- **Frontend Interface:** Single Page Application (SPA) with Vanilla JavaScript and Modern Responsive CSS (Segmented Mobile Tab Controls for phones and tablets)

5. REST API Endpoints Reference

Endpoint	Method	Payload	What It Does
/status	GET	None	Checks system health, loaded documents, page counts, chunk counts, and Groq status.
/upload	POST	multipart/form-data	Ingests a PDF, runs text/OCR extraction, splits into chunks, embeds, and stores in ChromaDB.
/ask	POST	JSON {question, model}	Performs semantic search, sends relevant context to Groq LLM, and returns answer + citations.
/tts	POST	JSON {text, voice}	Generates and streams an MP3 audio file of the answer for voice playback.
/delete-file	POST	JSON {filename}	Removes a specific PDF from disk and deletes its vector embeddings from ChromaDB.
/clear	POST	None	Clears all uploaded PDFs, resets the vector database, and wipes chat memory.
/clear-history	POST	None	Clears the chat conversation history without deleting your uploaded documents.
/update-config	POST	JSON {api_key, model}	Saves a new Groq API key or changes the active default model in runtime.

6. Quick Setup & Deployment Guide

Environment Configuration (.env file):

- GROQ_API_KEY = gsk_your_groq_key_here (Get free key at console.groq.com)
- GROQ_MODEL = llama-3.3-70b-versatile
- ELEVENLABS_API_KEY = optional_key (Optional for studio voice cloning)

Running Locally:

1. Install packages: `pip install -r requirements.txt`
2. Start server: `python app.py`
3. Open in browser: **http://127.0.0.1:5000**

Deploying on Render Cloud:

- **Build Command:** `pip install -r requirements.txt`
- **Start Command:** `gunicorn app:app --workers 1 --threads 4 --timeout 120`
- **Environment Variable:** Set GROQ_API_KEY in Render's Environment tab.